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I. Background

Murrelektronik Inc. is a leading manufacturer of industrial electronics. Industrial cables are their top selling product, and remain in high demand in the North American market. While most production takes place at the company's headquarters in Germany, cable production began in the United States in 2015. Target yield per day was at least 3000 cables, but the plant has repeatedly failed to meet set goal. When the plant cannot meet market demands cables have to be ordered from the German plant, at approximately \$0.50 higher cost¹. The production of cables starts with first cutting the cable to desired length, then adding required pins and head to the cable, and finally sealing the head in a plastic mold. Out of these three steps, cutting cable is the slowest process. As the first step, it bottlenecks the rest of the production. In this experiment, a new approach to cutting cable is being tested.



Picture 1: Sample of Murrelektronik Cables

II. Design

The intent of this experiment was to determine the effects of Cable-Cutting Method, Production Shift and the nuisance factor, Weekday (referred to as “Day”), on cable production output (Yield).

Design Components

Response: Yield, which is the total number of cables produced per shift.

Factor 1: Cable-cutting Method (referred to as Method), where the low level is the old method (Method 1), and the high level is the new method being tested (Method 2).

Factor 2: Production Shifts (referred to as Shift) of 8 hours each (Shift 1: 6:30 AM to 2:30 PM. Shift 2:

¹ To protect company privacy this is an entirely assumed figure and does not represent real cost differences.
4/14/2017

3:00 PM to 11:00 PM).

Block: Day of the week (referred to as Day) is treated as a nuisance factor (Monday = 1, and only weekdays are considered). Each day of the week is treated as its own block.

A balanced and replicated 2^2 Factorial Design with days as blocks was selected for this study. Figure 1 shows the breakdown of the design set up. Each cable-cutting method was tested three times per shift and per weekday (3 replicates). Sample size was pre-determined as shown below, using test runs standard deviation as an estimate for overall standard deviation, setting alpha level at 0.05 and desired power at 80%, along with expecting to detect a difference of at least 350 cables. This meant a total of 30 observations per method were needed, which required three weeks of runs per method.

$$n = 1 + (2 * 7.85) \left(\frac{467.90}{350} \right)^2 = 29.06 \approx 30$$

Figure 1: Design Replication

Factors	Blocks: Days of Week (1=Mon)					No. of Obs
	1	2	3	4	5	
Method 1	6	6	6	6	6	30
Shift 1	3	3	3	3	3	15
Shift 2	3	3	3	3	3	15
Method 2	6	6	6	6	6	30
Shift 1	3	3	3	3	3	15
Shift 2	3	3	3	3	3	15
No. of Obs	12	12	12	12	12	60

Next, factor levels were coded to allow determination of the relative importance of factors. For cable-cutting method, Method 1 was coded as “-1” and Method 2 was coded as “1”. For production shift, Shift 1 was coded as “-1” and Shift 2 was coded “1”. Figure 2 shows an excerpt of the design setup in Minitab².

Figure 2: Design Setup in Minitab

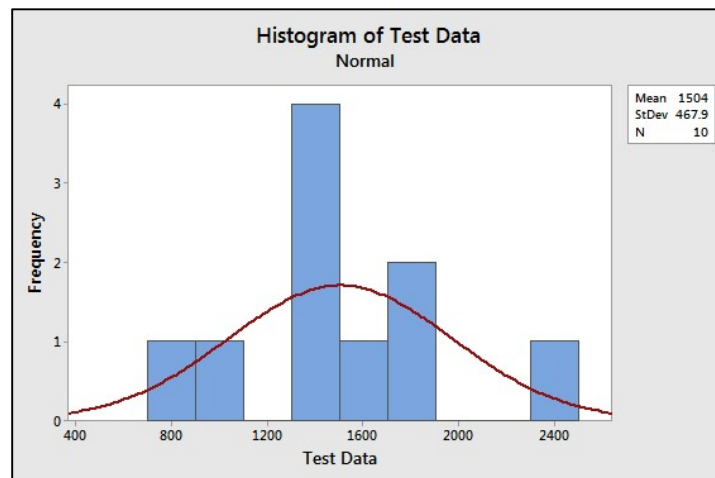
Run	Shift	Method	Day	Yield	StdOrder	CenterPt
1	-1	-1	2	1241	1	1
2	1	-1	2	1428	2	1
3	-1	-1	3	1875	3	1
4	1	-1	3	1327	4	1
5	-1	-1	4	1386	5	1
6	1	-1	4	1747	6	1
7	-1	-1	5	1656	7	1
8	1	-1	5	1365	8	1
9	-1	-1	1	1463	9	1
10	1	-1	1	1785	10	1

² Yield has been transformed to maintain company data privacy.

III. Measurement Evaluation

Cable output is automatically recorded in the companies SAP system along with the shift and date, which leaves no room for error in count. Figure 3 shows results from an initial run for Method 1 only. Mean for the initial 10 runs was 1504 cables (Yield) with a standard deviation of 467.9. Measurements are consistent with all observations falling within ± 2 standard deviations.

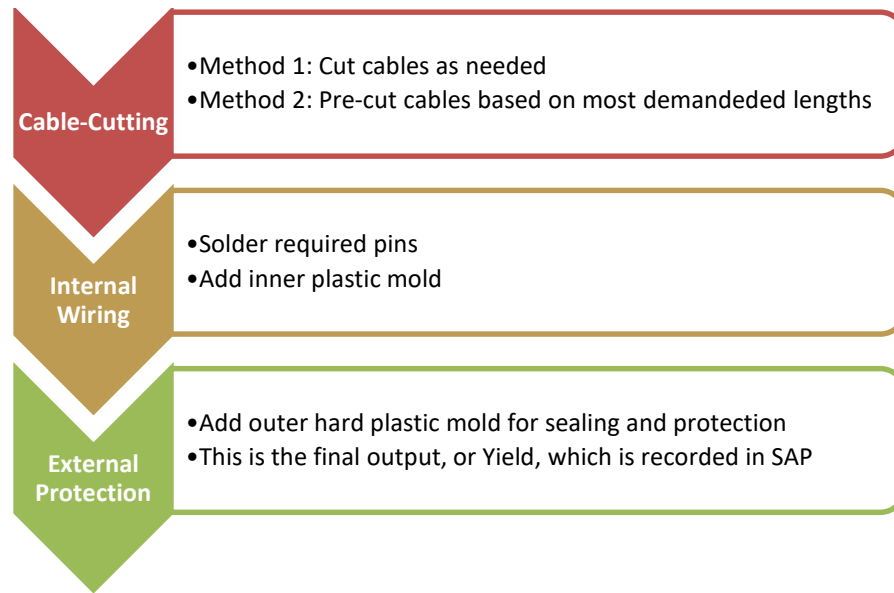
Figure 3: Spread of Test Run for Method 1



IV. Data Collection Plan

Figure 4 gives a brief step-by-step breakdown of the production process, along with some notes on each step in the corresponding bubble. Day of the week will be extracted from the date. Cable-cutting Method 2 is set to be introduced on 2/23/17 unbeknownst to the production team handling the latter part of production. Only the cable cutters are informed of the change. This “blinding” is added to eliminate bias. Randomization of cable-cutting method was not selected in order to avoid confusion and minimize errors on the production floor. Since randomization is difficult blocking will instead be used to make detection of true effects more likely. Counts of each step in production are recorded, but only final cable output is considered for this design.

Figure 4: Cable Production Process Overview



V. Results

The results from the experiment are summarized in Figure 5, which shows the averages for the three replicates for each factor and block. Method 1 average is 1940 cables whereas Method 2 average is 2516 cables, so there does seem to be an improvement in Yield over the two methods. Production shift, on the other hand, doesn't seem to vary a lot within the two different methods. However, to obtain conclusive and statistically valid results an ANOVA test must be conducted.

Figure 5: Average Yields across the Experiment

Factors	Blocks: Days of Week (1=Mon)					No. of Obs
	1	2	3	4	5	
Method 1	2044	1635	1983	1977	2060	1940
Shift 1	1985	1434	2184	1916	2114	1926
Shift 2	2102	1836	1782	2037	2006	1953
Method 2	2309	2282	2480	2735	2776	2516
Shift 1	2408	2040	2710	2491	3027	2535
Shift 2	2210	2523	2250	2978	2525	2497
No. of Obs	2176	1958	2231	2356	2418	

Data Analysis: Figure 6 shows the effect sizes for all factors and their interactions. Cable-cutting method has the highest effect on this model with an effect size of 576.70. In comparison, Shift and Shift*Method interaction have a very low effect size. Block 2 (Tuesday) has a highly negative coded coefficient indicating productivity issues on Tuesdays. Figure 7 shows the Pareto Chart for the initial model, which helps visualize how much of an impact Method has on cable yield compared to Shift and their interaction.

Figure 6: Factor Effects

Coded Coefficients						
Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		2227.9	54.8	40.66	0.000	
Blocks						
1		-52	110	-0.47	0.639	1.60
2		-270	110	-2.46	0.017	1.60
3		3	110	0.03	0.975	1.60
4		128	110	1.17	0.249	1.60
Shift (A)	-5.8	-2.9	54.8	-0.05	0.958	1.00
Method(B)	576.7	288.4	54.8	5.26	0.000	1.00
Shift*Method	-31.9	-16.0	54.8	-0.29	0.772	1.00

Figure 7: Pareto Chart for Effects on Yield

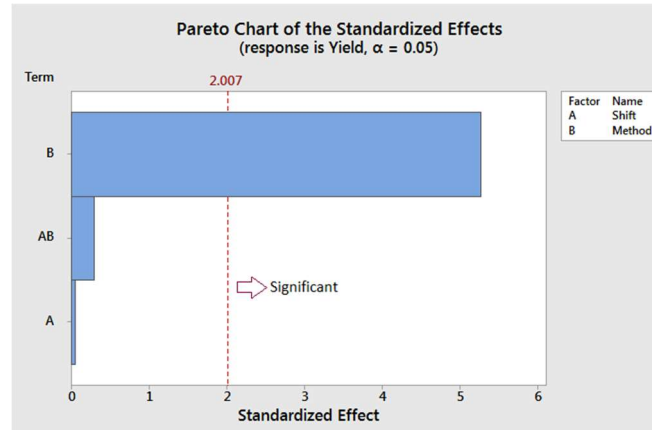


Figure 8 shows the ANOVA results for the initial model. Only Method is significant with a p-value of 0.000 at $\alpha = 0.05$. Shift is highly insignificant with a p-value of 0.958. Blocking on Days is also insignificant with a p-value of 0.09. Thus, another model without blocking was also explored. The results didn't change for the factors, but the model without the blocks lost R-Squared slightly, so the original model with the blocking was kept since it reduces experimental error and helps detect true factor effects when randomization is not possible.

Figure 8: ANOVA Results for Initial Model

Factorial Regression: Yield versus Day, Shift, Method					
Analysis of Variance					
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	7	6540401	934343	5.19	0.000
Blocks	4	1535280	383820	2.13	0.090
Linear	2	4989825	2494912	13.85	0.000
(A) Shift	1	505	505	0.00	0.958
(B) Method	1	4989320	4989320	27.70	0.000
2-Way Interactions	1	15296	15296	0.08	0.772
Shift*Method	1	15296	15296	0.08	0.772
Error	52	9366622	180127		
Lack-of-Fit	12	2469791	205816	1.19	0.321
Pure Error	40	6896831	172421		
Total	59	15907023			
Model Summary					
S	R-sq	R-sq(adj)	R-sq(pred)		
424.414	41.12%	33.19%	21.60%		

However, since the interaction wasn't significant another model was run (shown in Figure 9), which excluded the interaction term. This slightly improved the R-squared (adj.) from 33.19% to 34.34%. This also provides a simpler regression equation. The equation is averaged over blocks. The intercept of 2227.9 (or 2228 cables) is the overall mean of Yield. Since Method 2 is coded as "1", it can increase output by 288.4 (or 289 cables) per day and per shift. Whereas Method 1 decreases total Yield by 289 cables per day and per shift. Shift, on the other hand, has a very small effect on total Yield.

Figure 9: ANOVA Results for Refined Model

Factorial Regression: Yield versus Day, Shift, Method						
Analysis of Variance						
Source	DF	Adj SS	Adj MS	F-Value	P-Value	
Model	6	6525105	1087517	6.14	0.000	
Blocks	4	1535280	383820	2.17	0.085	
Linear	2	4989825	2494912	14.09	0.000	
(A) Shift	1	505	505	0.00	0.958	
(B) Method	1	4989320	4989320	28.19	0.000	
Error	53	9381919	177017			
Lack-of-Fit	13	2485087	191161	1.11	0.380	
Pure Error	40	6896831	172421			
Total	59	15907023				
Model Summary						
	S	R-sq	R-sq(adj)	R-sq(pred)		
	420.734	41.02%	34.34%	24.41%		
Coded Coefficients						
Term	Effect	Coef	SE Coef	T-Value	P-Value	VIF
Constant		2227.9	54.3	41.02	0.000	
Blocks						
1		-52	109	-0.48	0.636	1.60
2		-270	109	-2.48	0.016	1.60
3		3	109	0.03	0.975	1.60
4		128	109	1.18	0.245	1.60
Shift	-5.8	-2.9	54.3	-0.05	0.958	1.00
Method	576.7	288.4	54.3	5.31	0.000	1.00
Regression Equation in Uncoded Units						
Yield = 2227.9 - 2.9 Shift + 288.4 Method						

Next, to ensure the validity of ANOVA results a residual analysis was conducted. Figure 10 shows the residual plots for Yield. Only the Independence of Runs plot possibly violates the assumptions, but this doesn't seem severe.

- Figure 10 shows the residuals are normally distributed with a p-value of 0.29 for the AD test.

- Residuals vs Fitted Value Plot has a random pattern indicating homogenous variances. Also, Levene's test in Figure 11 has a p-value of 0.999, which indicates homogenous variances. Factor plots vs. residuals in Figure 12 exhibit a balanced and random pattern as well.
- The histogram shows that all residuals are within ± 2 standard deviations, thus outliers are not a concern.
- The run order plot is unbalanced about 0 before the 12th run. This could be due to no randomization of cable-cutting method. However, Method 1 spans runs 1 through 30, and residuals by run order do seem randomly distributed around zero after the 13th run.

Figure 10: Residual Analysis for Refined Model

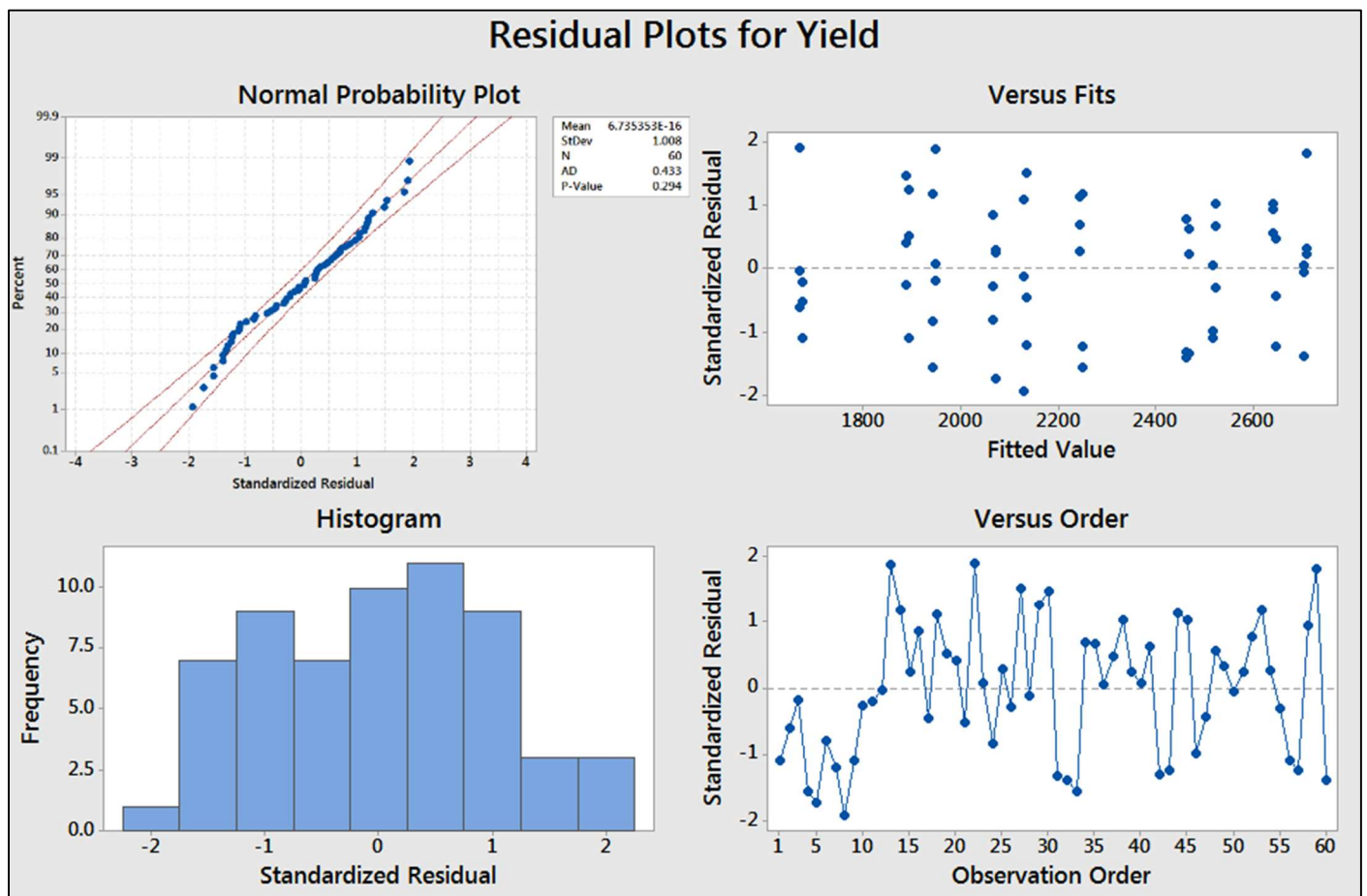
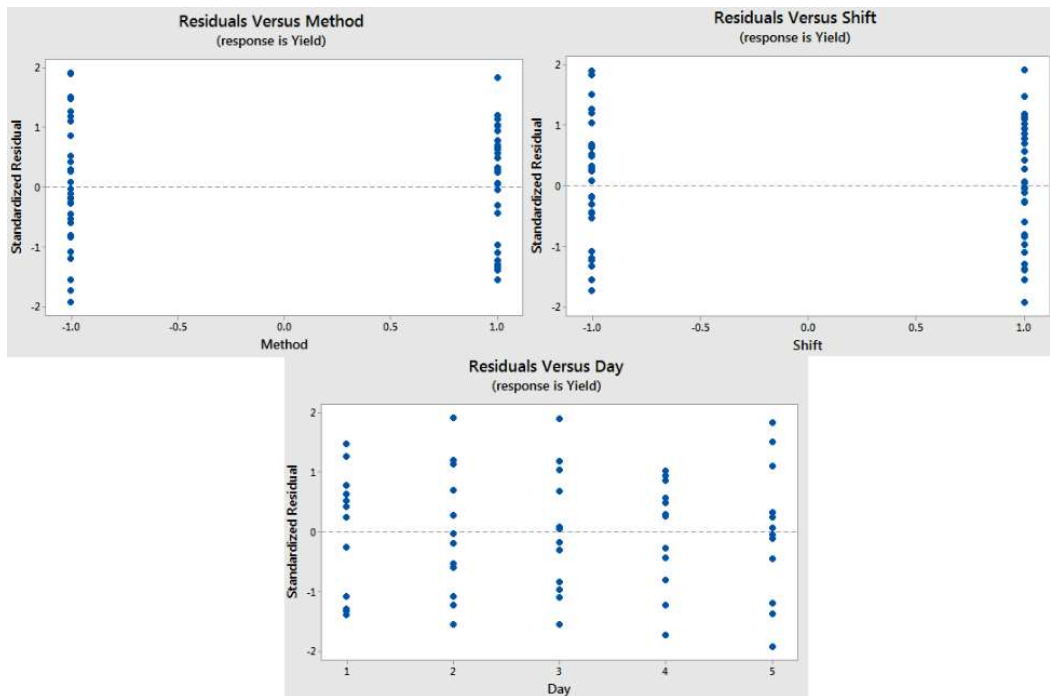


Figure 11: Levene's Test for Homogenous Variances

Tests		
Method	Test Statistic	P-Value
Multiple comparisons	-	0.773
Levene	0.25	0.999

Figure 12: Factor and Block Plots vs Residuals



From the Main Effects Plot shown in Figure 13 it can be seen that Shift 1 vs. Shift 2 has no significant effect on cable yield. Whereas, Method 2 (coded 1) greatly increases cable yield compared to Method 1 (coded -1). The block effects can also be seen indicating more productivity later in the week (Thursday to Friday). Furthermore, Figure 14 shows the Tukey Pairwise Comparison for Shift and Method. Only Method yields significantly different groupings in means.

Figure 13: Main and Block Effects for Yield

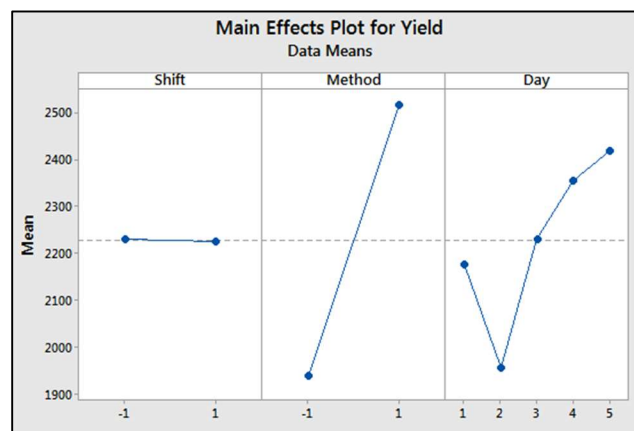


Figure 14: Tukey Pairwise Comparisons

Tukey Pairwise Comparisons: Response = Yield, Term = Shift				
Grouping Information Using the Tukey Method and 95% Confidence				
Shift	N	Mean	Grouping	
-1	30	2230.8	A	
1	30	2225.0	A	

Tukey Pairwise Comparisons: Response = Yield, Term = Method				
Method	N	Mean	Grouping	
1	30	2516.27	A	
-1	30	1939.53	B	

Means that do not share a letter are significantly different.

VI. Conclusion

The results indicate that the new cable-cutting method is a great step towards increasing cable yield. It also shows that both shifts work equally hard, and have no significant differences in cable yield. The nuisance factor “Days” does not have a significant effect on cable output. Figure 15 shows a contour plot for Method vs Shift, and indicates high yield output for Method 2 (coded 1), but has the same gradient for Shift 1 vs. 2 indicating no change on yield. Figure 16 shows how the regression equation from Figure 9 can be used to maximize cable yield by selecting Method 2 (coded 1) and Shift 1 (coded -1). Finally, Figure 17 summarizes how Murrelektronik can increase cable yield by using Method 2 over Method 1, and how this will on average save them \$288 a day and about \$103K annually.

Figure 15: Contour Plot for Yield vs Factors

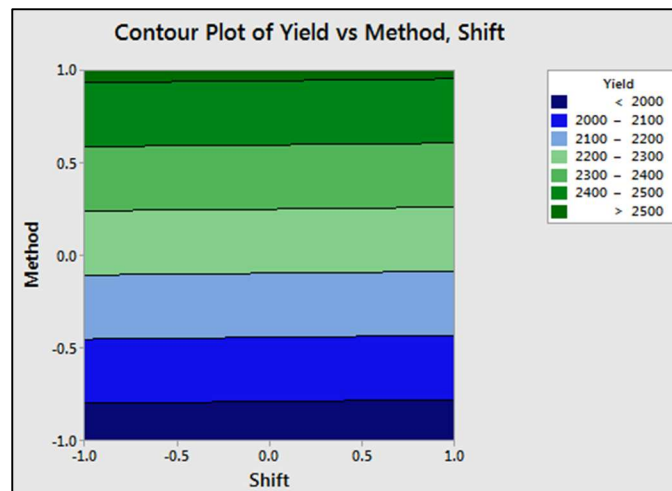


Figure 16: Optimum Level Selection Chart



Figure 17: Summary of Potential Savings

	A	B	Increase in Yield	
Produced	1940	2516	30%	
Target	3000	3000	Savings	
Difference	1060	484	\$ 288.00	Daily
Reduced Cost	\$ 530.00	\$ 242.00	\$ 8,640.00	Monthly
			\$ 103,680.00	Annually